

Unit: 20

AC CIRCUIT**ALTERNATING CURRENT:****Definition:**

“The electric current whose magnitude changes continuously and direction reverses periodically is called alternating current”.

Explanation:

Alternating current (A.C) is that which is produced by a voltage source whose polarity keeps on reversing with time. The magnitude of alternating current changes continuously with time and its direction is reversed periodically. It is represented by a sine curve or a cosine curve.

$$I = I_0 \sin \omega t$$

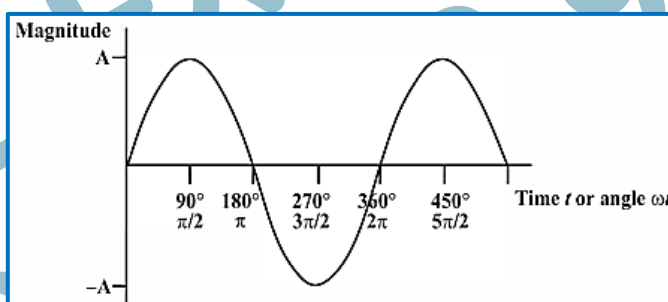
Where “I” is Instantaneous value and I_0 is peak value of current.

Graphically:

A graphical representation shows the nature of an alternating current (as shown in Figure).

This graph shows that during one half of the cycle, the current is positive, while during the other half, it is negative.

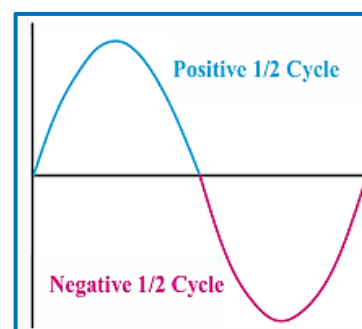
Consequently, the current alternates its direction in the wires it traverses, flowing first in one direction and then in the opposite direction.

**AC TERMINOLOGIES:****1. CYCLE:**

One complete set of both positive and negative values of an alternating quantity is known as a cycle (as shown in figure).

Alternating quantity is positive for first half of the cycle and negative for the rest half.

Hence average value of alternating quantity (i or V) over a complete cycle is zero.

**2. TIME PERIOD (T):**

The time taken of an alternating current (AC) or voltage wave to complete one cycle of variations is called the periodic time or time period.

It is denoted by the symbol T.

The relationship between time period and frequency (f) is given by the equation: $T = 1/f$. Where T is in second.

3. FREQUENCY (F):

The number of complete cycles of the current waveform that occur in one second is called frequency of alternating current (AC) or voltage wave.

Frequency is a measure of how many times the direction of the current reverses per second.

It is typically expressed in hertz (Hz), where one hertz equals one cycle per second.

The relationship between frequency and time period is expressed as $f = 1/T$

4. INSTANTANEOUS VALUE:

Definition:

The value of voltage or current at any instant of time, measured from some reference point, is known as its instantaneous value.

Explanation:

It is denoted by V and can have any value between plus maximum value $+V_0$ and negative maximum value $-V_0$. The entire waveform (as shown in figure) is actually a set of all the instantaneous values that exist during a period, T .

Mathematically, it is given by $V = V_0 \sin \theta = V_0 \sin \omega t$

5. PEAK VALUE:

Definition:

"The maximum or highest value of alternating quantity (current or Voltage) is defined as peak value or amplitude."

Explanation:

For a sine wave, this is the highest positive or negative value reached during a cycle. It is usually denoted as I_0 for current and V_0 for voltage.

6. ROOT MEAN SQUARE (RMS) VALUE:

Definition:

"Square root of the average of squares of instantaneous values of alternating current or voltage in one cycle is called root mean square value."

Explanation:

The root mean square (RMS) value of AC represents its equivalent DC value for producing the same heat in a resistor.

The average value of voltage or current is not used in electric power calculations. The reason being that the AC cycle consists of positive and negative half cycles, so the average value is zero.

$$V_{av} = 0$$

However, there exists a mathematical relation between the peak value V_0 of alternating voltage and a direct current (DC) voltage that yields an equivalent average electrical power. This constant DC voltage is referred to as the root-mean-square (r.m.s) value of the alternating voltage V_{rms} .

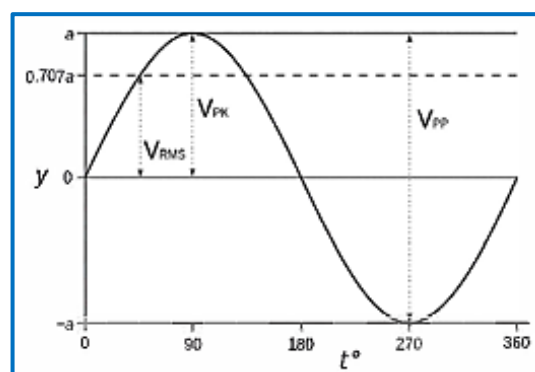
The RMS value of a sinusoidal AC waveform is equal to its peak value divided by the square root of two.

$$V_{rms} = \frac{V_0}{\sqrt{2}}$$

The DC voltage equivalent to rms voltage is approximately 70% of V_{peak} or V_0 as shown in figure.

For a sinusoidal wave, the rms value is approximately 0.707 times the peak value.

If we connect an ordinary D.C. ammeter to measure alternating current, it will measure its value as average over a cycle.



Since the average value of current and voltage over a cycle is zero, but the power delivered during a cycle is not zero because power is I^2R and the values of I^2 are positive even for negative values of " I ".

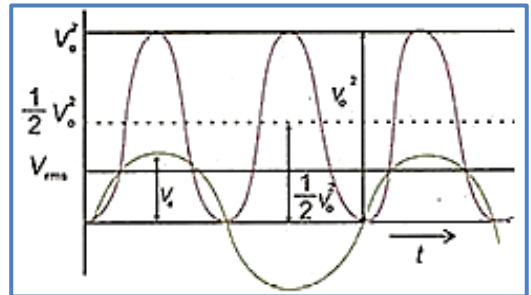
Thus, the average value of I^2 is not zero and is called the mean square current.

The alternating current or voltage is measured by square root of its mean square value, known as root mean square (rms) value.

Method of Finding rms value:

Let us compute the average value of V^2 over a cycle. Figure shows an alternating voltage and the way its V^2 values vary. The values of V^2 are positive on the negative half cycle also.

As the graph of V^2 is symmetrical about the line $\frac{1}{2} V_0^2$ so for this figure, the mean or the average value of V^2 is $\frac{0^2 + V_0^2}{2} = \frac{1}{2} V_0^2$. The root mean square (rms) value of V is obtained by taking the square root of $\frac{V_0^2}{2}$. Therefore



$$V_{rms} = \sqrt{\frac{0^2 + V_0^2}{2}}$$

$$V_{rms} = \sqrt{\frac{V_0^2}{2}}$$

$$V_{rms} = \frac{V_0}{\sqrt{2}}$$

$$V_{rms} = 0.707 V_0$$

Similarly,

$$I_{rms} = 0.707 I_0$$

Most of the alternating current and voltage meters are calibrated to read rms values. When we speak of A.C meter reading, we usually mean rms values unless stated otherwise.

ALTERNATING CURRENT OR VOLTAGE EQUATION:

When a coil undergoes constant rotation at a uniform angular velocity within a magnetic field, it generates an alternating voltage.

Alternating Current (AC) or Voltage refers to the type of electrical flow or potential difference that periodically changes direction and magnitude over time.

The fundamental equation describing AC voltage or current in a sinusoidal waveform is:

$$V = V_0 \sin \omega t \dots\dots\dots (1)$$

$$I = I_0 \sin \omega t \dots\dots\dots (2)$$

Where: "

V ", " I " represent the instantaneous value of the voltage and alternating current respectively.

V_0 , I_0 is the amplitude of the waveform, which represents the maximum value of alternating voltage and alternating current respectively.

ω is the angular frequency, given by $\omega = 2\pi f$

WORKED EXAMPLE:

20.1: If an alternating current flows through a circuit with an amplitude of 5 amps, with frequency of 60 Hz, write its equation.

PHASE OF AC:

Definition:

"The angle which represents the instantaneous value of alternating current or voltage is known as its Phase."

Explanation:

The phase describes the relative positions of the waveforms of alternating values. Hence in alternating current (A.C) circuits, the concept of phase refers to the relationship between different waveforms as time passes.

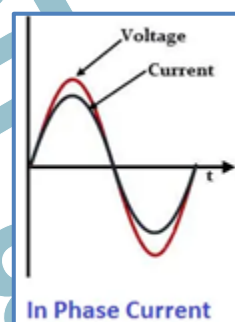
If an alternating quantity is expressed as " $v = V_0 \sin(\omega t \pm \phi_0)$ ", then angle $(\omega t + \phi)$ is called its phase.

Where ωt = instantaneous phase (changes with time) and ϕ_0 = initial phase (constant w.r.t time).

STATES OF PHASE WITHIN AC CIRCUIT:

1. IN-PHASE:

When two waveforms (have the same frequency) reach their peak values or zero values at the same time, they are said to be in phase.

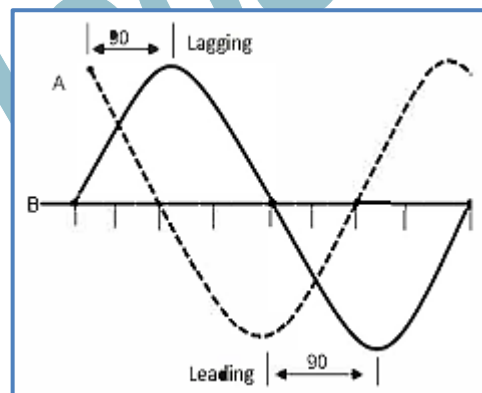


2. PHASE LAG:

If one waveform reaches its peak or zero value after the other, the second waveform is said to lag the first.

It is shown by figure, where a wave shown by solid line lags the wave shown by dotted line.

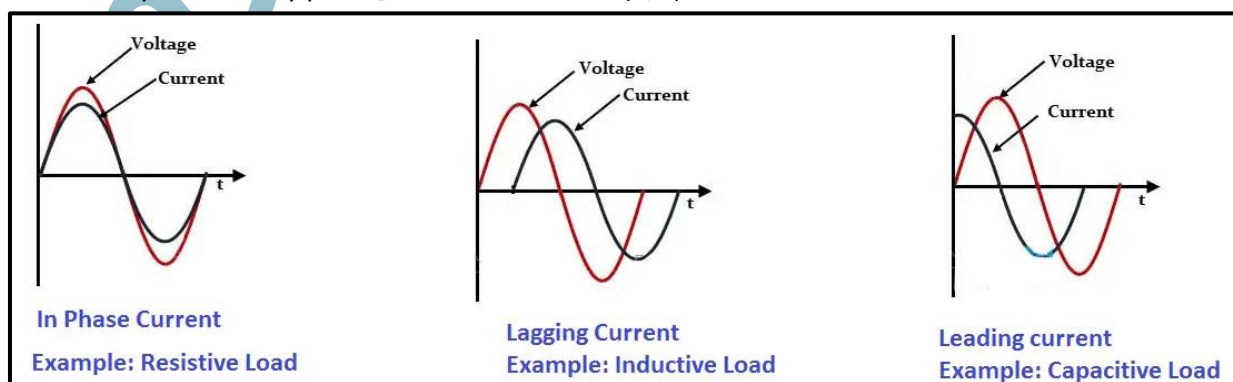
It can be said that the solid wave form lags the dotted line waveform by 90° or $(\pi/2)$ radians.



3. PHASE LEAD:

If one waveform reaches its peak or zero value before the other, the first waveform is said to lead to the second.

It is shown by figure where a wave shown by dotted line leads the wave shown by solid line by phase difference of 90° or $(\pi/2)$ radians.



PHASOR:

Definition:

The representation of phase lead and lag between two alternating quantities is effectively demonstrated by showing the two AC quantities as vectors also known as "Phasor". OR

"A rotating vector used to represent a sinusoidal waveform is called Phasor."

It's a dimensionless quantity and its unit is radian.

Explanation:

A sinusoidally alternating voltage or current can be graphically represented by a counterclockwise rotating vector provided it satisfies the following conditions.

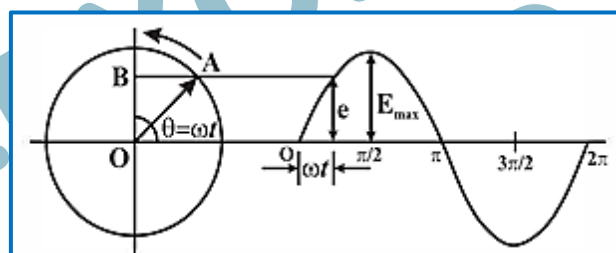
1. Its length on a certain scale represents the peak or rms value of the alternating quantity.
2. It is in the horizontal position at the instant when the alternating quantity is zero and is increasing positively.
3. The angular frequency of the rotating vector is the same as the angular frequency " ω " of the alternating quantity.

VECTOR REPRESENTATION OF AN ALTERNATING QUANTITY:

To solve AC problems effectively, it is beneficial to represent a sinusoidal quantity, such as voltage or current, using a line of specific length that rotates in a counterclockwise direction with the same angular velocity " ω " as the sinusoidal quantity. This rotating line is commonly referred to as a phasor.

Let's consider a line denoted as OA, referred to as a phasor, which accurately represents, to scale, the maximum value of an alternating quantity, such as electromotive force (emf).

In this representation, OA equals the maximum emf value and rotates counterclockwise at an angular velocity of ω radians per second around point O, as shown in Figure.



An arrowhead is placed at the outer end of the phasor, serving both to indicate the assumed direction of movement and to specify the precise length of the phasor, especially when multiple phasors coincide.

In Figure, OA represents the phasor after it has rotated through an angle θ , equivalent to " ωt ", from its initial position when the emf was at its zero value.

The projection of OA on the Y-axis, denoted as OB, equals OA multiplied by $\sin\theta$, which is equivalent to $E_{\max}\sin\omega t$, representing the instantaneous value of the emf, denoted as 'e,' at that moment.

Therefore, the projection of OA on the vertical axis accurately portrays, to scale, the instantaneous value of the emf.

It means that a **phasor** is a **rotating vector** (line) in the complex plane. It has:

- A **length** equal to the amplitude of the sinusoidal wave.
- A **direction** (angle) that represents the phase of the wave.
- It **rotates counterclockwise** at the same angular velocity ω as the sinusoidal signal.

PHASE DIFFERENCE (PHASE CONSTANT):

The difference between the phases of currents and voltage is called phase difference.

If alternating voltage and current are given by $V = V_0 \sin(\omega t + \phi_1)$ and $I = I_0 \sin(\omega t + \phi_2)$ then phase difference $\phi = \phi_1 - \phi_2$ (relative to current) or $\phi = \phi_2 - \phi_1$ (relative to voltage)

PHASOR DIAGRAM:

A diagram representing alternating current and alternating voltage of same frequency as vectors (phasors) with the phase angle between them is called a phasor diagram.

While drawing phasor diagram for a pure element (e.g. R, L or C) either of the current or voltage can be plotted along X-axis.

But when phasor diagram for a combination of elements is drawn then quantity which remains constant for the combination must be plotted along X-axis, so we observe that

- (i) In series circuits current has to be plotted along X-axis.
- (ii) In parallel circuits voltage has to be plotted along X-axis.

SELF-ASSESSMENT QUESTIONS:

1. Explain the concept of reactance in AC circuits and how it differs from resistance.
2. How does the phase angle affect the relationship between voltage and current in AC circuits?

SIMPLE AC CIRCUITS:

The basic circuit element in a D.C. circuit is a resistor (R) which controls the current or voltage and the relationship between them is given by Ohm's law that is $V = IR$.

In A.C. circuits, in addition to resistor R, two new circuit elements, namely inductor (L), and capacitor (C), are relevant. The current and voltages in A.C. circuits are controlled by three elements R, L and C.

1. AC THROUGH A RESISTOR:

Consider a circuit consisting of a pure resistor (R), connected to an alternating voltage source, as shown in figure. At any given time 't,' the potential difference across the resistor's terminals is expressed as

$$V = V_0 \sin \omega t \dots\dots\dots (1)$$

Where V_0 is the peak value of the alternating voltage.

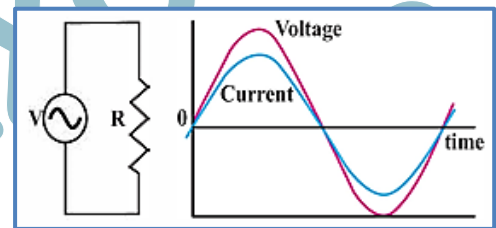
The circuit's current (I) is governed by Ohm's law:

$$\begin{aligned} I &= \frac{V}{R} \\ &= \frac{V_0 \sin \omega t}{R} \\ &= \frac{V_0}{R} \sin \omega t \end{aligned}$$

Or $I = I_0 \sin \omega t \dots\dots\dots (2)$

The current achieves its maximum value when $\sin \omega t = 1$.

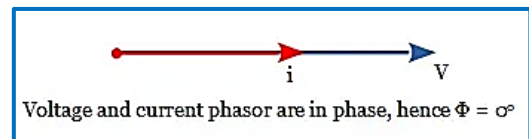
From equations (1) and (2), it is evident that the applied voltage and the current in the circuit are in phase with each other as shown in figure (b).



PHASOR DIAGRAM OF RESISTIVE CIRCUIT:

The phase relationship is further illustrated by the phasor diagram as shown in figure.

The current (I) flowing through resistor (R) is in phase with the voltage (V) across the resistor. This alignment can be visually represented on a phasor diagram by depicting a vector that coincides with the voltage (V).



The power loss in a resistor in an AC circuit is a result of the conversion of electrical energy into heat due to the resistance of the resistor.

The instantaneous power in the resistance can be expressed using the formula.

$$P = I^2 R \quad \text{Where, } P = V_{\text{rms}} \cdot I_{\text{rms}}$$

WORKED EXAMPLE:

- 20.2: A 2000 resistor is connected with AC supply of 36V with frequency of 60 Hz. Calculate (i) Current flowing through the circuit. (ii) Power dissipated in the resistor. (iii) Also write equations for current and voltage as a function of time.

2. A.C THROUGH CAPACITORS:

BEHAVIOR OF CAPACITOR IN DC SOURCE:

When capacitor is connected to a direct current (DC) supply voltage, then capacitor plates gradually accumulate charge until the voltage across the capacitor matches with the source voltage. The capacitor retains this charge as a temporary storage device if the applied voltage is sustained.

The ability of a capacitor to store charge on its plates is termed as capacitance (C).

During the charging process, an electric current flows into the capacitor, causing its plates to acquire an electrostatic charge. This charging current is highest when the capacitor plates are uncharged, decreasing exponentially over time until the capacitor reaches full charge.

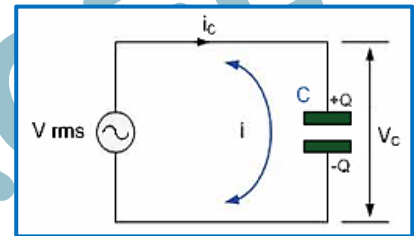
Once the capacitor is fully charged, it blocks further electron flow onto its plates as they become saturated.

Hence it is often said that capacitor blocks direct current after initial transient time.

BEHAVIOR OF CAPACITOR IN AC SOURCE:

Now consider a capacitor connected to the AC source as shown in the figure.

- During the positive half cycle of alternating voltage, the electrons flow from upper capacitor plate to the source, leaving it as positively charged “+Q” and source supplies equal number of electrons to the lower plate making it as negatively charged “-Q” as shown in figure.
- During the negative half cycle of alternating voltage, the direction of motion of electrons is also reverted resulting in the capacitor plates becoming charged in the opposite manner and current flows in opposite direction, hence capacitor charges and discharges. In this way the alternating current flows through the capacitor.



Mathematically:

If alternating source voltage ($V = V_0 \sin \omega t$) is applied to the capacitor, the charge on any plate of capacitor is given by $q = CV$

By putting the value of V ($V = V_0 \sin \omega t$), we get

$$q = C V_0 \sin \omega t \dots\dots\dots (1)$$

Since C and V_0 are constants, it is obvious that q will vary the same way as applied voltage. From equation (1) q and V are in phase.

The current “ I ” is the rate of change of q with time i.e.,

$$I = \frac{\Delta q}{\Delta t} \dots\dots\dots (2)$$

Substitute equation (1) in (2), we get

$$I = \frac{C V_0 \Delta \sin \omega t}{\Delta t}$$

$$I = C V_0 \cdot \frac{\Delta \sin \omega t}{\Delta t}$$

$$\text{OR } I = C V_0 \omega \cos \omega t$$

Since $\cos \omega t = \sin \left(\omega t + \frac{\pi}{2} \right)$. Therefore

$$I = C V_0 \omega \sin \left(\omega t + \frac{\pi}{2} \right) \dots\dots\dots (3)$$

Multiply and divide the right side of equation (3) by ωC

$$I = \frac{V_0 \sin \left(\omega t + \frac{\pi}{2} \right)}{\frac{1}{\omega C}} \dots\dots\dots (4)$$

Where $\frac{1}{\omega C} = X_c$ is the capacitive reactance. Therefore equation (4) becomes

$$I = \frac{V_0 \sin \left(\omega t + \frac{\pi}{2} \right)}{X_c}$$

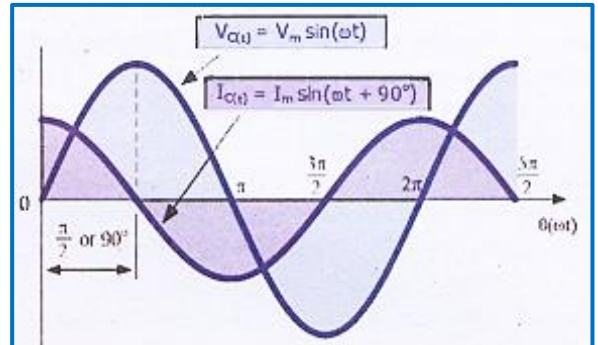
$$I = \frac{V_o}{X_c} \sin\left(\omega t + \frac{\pi}{2}\right) \quad \left(\text{Since } \frac{V_o}{X_c} = I_o\right)$$

Therefore $I = I_o \sin\left(\omega t + \frac{\pi}{2}\right) \dots\dots (5)$

Equation (5) shows that in pure capacitive circuits the current exhibits sinusoidal variation and it leads the voltage, by 90 degrees or $(\pi/2)$ radians or the voltage lags the current by 90 degrees or $(\pi/2)$ radians as shown in figure.

The reason is that when a voltage is introduced to an initially uncharged capacitor, the capacitor exhibits low reactance or impedance, resulting in the maximum current draw.

As the capacitor becomes charged, the current diminishes (zero), causing the voltage across the capacitor to increase. Once the capacitor is fully charged, there is no further current flow, and the voltage attains its maximum level, hence voltage lags the current by 90° .



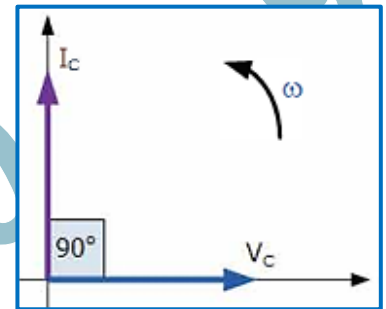
PHASOR DIAGRAM OF CAPACITIVE CIRCUIT:

The phasor diagram is a graphical tool that illustrates the relationship between voltage and current in a capacitor using vectors, known as phasors.

Since, in an AC circuit, a capacitor experiences a phase shift between the voltage and the current.

In the phasor diagram shown in Figure, " V_c " represents the voltage across the capacitor, while " I_c " denotes the current flowing through it.

The angle $\phi = 90^\circ$ indicates the phase difference by which the current leads the voltage in a capacitor.



REACTANCE OF CAPACITOR:

Definition:

Reactance of a capacitor is a measure of the opposition offered by the capacitor to the flow of A.C.

Explanation:

In a purely capacitive circuit, capacitive reactance represents the opposition to alternating current flow.

Similar to resistance, reactance is measured in Ohms, but it is denoted by the symbol X_c to differentiate it from purely resistive values.

Since $I = \frac{V}{X_c}$

From equation (4) $(I = \frac{V_o}{\frac{1}{\omega C}} \sin(\omega t + \frac{\pi}{2}))$, we get.

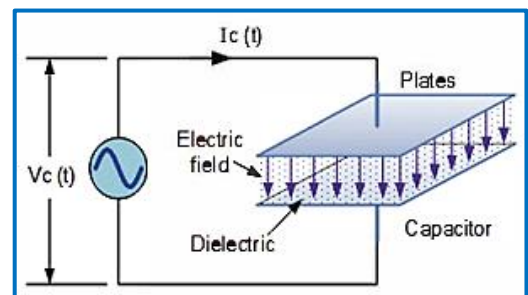
$$I_o = \frac{V_o}{\frac{1}{\omega C}} \dots\dots\dots (5)$$

Like ohms law ($I = \frac{V}{R}$) the term in denominator of equation (5) represents the opposition to flow of current; in this case it is called reactance of capacitor.

Hence $X_c = \frac{1}{\omega C}$ (Since, $\omega = 2\pi f$)

So $X_c = \frac{1}{2\pi f C} \dots\dots\dots (6)$

The capacitive reactance shows an inverse relationship with the frequency of the applied alternating voltage. For lower frequencies, the reactance of capacitors increases, while at high frequencies, the reactance decreases.



SELF-ASSESSMENT QUESTIONS:

1. Why does the root mean square (RMS) value used to represent AC voltage and current instead of instantaneous peak values?
2. How does the phase angle affect the relationship between voltage and current in AC circuits?

WORKED EXAMPLE:

- 20.3: A 200Ω resistor is connected with AC supply of 36V with frequency 60 Hz. Calculate (i) Current flowing through the circuit (ii) Power dissipated in the resistor. (iii) Also write equations for current and voltage as a function of time.

3. A.C THROUGH INDUCTOR:**INDUCTOR:**

An inductor is a passive electrical component which can be formed by winding a conducting wire over an insulating object, such as pencil.

An inductor is usually in the form of a coil, or a solenoid wound from a thick wire so that it has a large value of self-inductance and has negligible resistance.

Example:

Solenoid is an example of inductor.

BEHAVIOR OF INDUCTOR IN AC SOURCE.

The primary purpose of an inductor is to oppose changes in current. It resists the flow of alternating current (AC) and stores energy in its magnetic field during the on-time of the AC cycle and releases it during the off-time.

Consider an inductor in the form of solenoid connected with A.C source as shown in figure.

When the switch is closed, current starts to flow through the inductor, the magnitude and direction of current is changing hence associated magnetic field also varies due to which an induced emf is set up in the inductor so as to oppose the change in accordance with the Lenz law.

According to Faraday's law, the magnitude of induced emf is given by.

$$\varepsilon = L \frac{\Delta I}{\Delta t}$$

To maintain the current, the applied voltage must be equal to the back e.m.f.

Therefore, the magnitude of voltage across the coil must be equal to induced or back emf. i.e. $V = \varepsilon$

$$V = L \frac{\Delta I}{\Delta t} \dots\dots\dots (1)$$

The alternating voltage produces a sinusoidal current, that is $I = I_0 \sin \omega t$. Hence equation (1) becomes.

$$V = L \frac{I_0 \Delta \sin \omega t}{\Delta t}$$

$$V = L I_0 \left(\frac{\Delta \sin \omega t}{\Delta t} \right)$$

Or

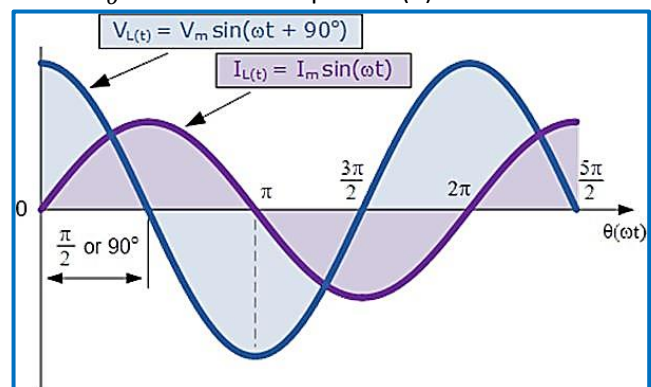
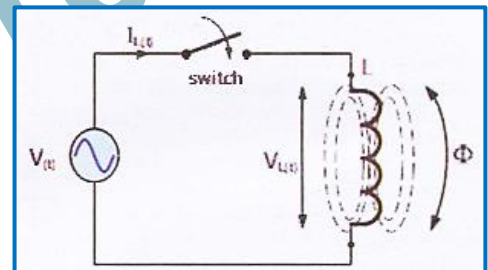
$$V = \omega L I_0 \cos \omega t$$

Where $\omega L I_0 = V_0$, therefore

$$V = V_0 \cos \omega t \dots\dots\dots (2)$$

Using $\cos \omega t = \sin \left(\omega t + \frac{\pi}{2} \right)$. Equation (2) becomes

$$V = V_0 \sin \left(\omega t + \frac{\pi}{2} \right) \dots\dots (3)$$



From equation (3) it is clear that in case of inductors connected in A.C circuit the voltage leads the current by $(\pi/2)$ radians as shown in figure.

The reason for voltage leading in an inductive circuit is due to the generation of induced electromotive force (EMF), commonly known as back EMF, when an alternating voltage is applied to the inductor.

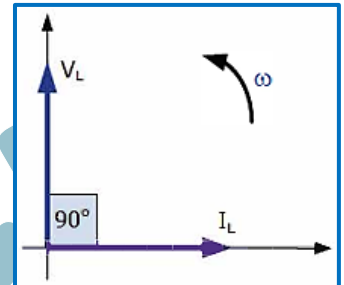
PHASOR DIAGRAM OF INDUCTIVE REACTANCE:

This back EMF appears instantaneously and induces a counter-current flow, introducing a time delay, typically in the order of milliseconds.

As a result, there is a brief time lag for the circuit current to overcome this opposing current and attain its maximum value.

That is why the voltage manifests first followed by the appearance of the current after a short interval.

Hence the voltage leads the current or the current lags behind the voltage in an inductive AC circuit, draw I_L behind V_L by a phase angle of 90 degrees as shown by phasor diagram in figure.



INDUCTIVE REACTANCE:

Definition:

The opposition to the flow of alternating current by an inductor is called inductive reactance.

It is denoted by X_L . It is measured in ohms.

Explanation:

In a purely inductive circuit, inductive reactance represents the opposition to alternating current flow.

Like resistance, reactance is measured in Ohms, but it is denoted by the symbol X_L to differentiate it from purely resistive values.

Since
$$I = \frac{V}{X_L} \dots\dots\dots (5)$$

From equation ($V = \omega L I_o \cos \omega t$), we get.

$$V_o = \omega L I_o$$

$$I_o = \frac{V_o}{\omega L} \dots\dots\dots (6)$$

By comparing equation (5) and equation (6), we get

Hence
$$X_L = \omega L \quad \quad \quad (\text{Since, } \omega = 2\pi f)$$

So
$$X_L = 2\pi f L \dots\dots\dots (7)$$

Where: X_L is the inductive reactance, “f” is the frequency of the AC signal, “L” is the inductance of the inductor.

Equation (4) shows that inductive reactance is directly proportional to the frequency of the AC signal.

As the frequency increases, the inductive reactance also increases.

CHOKE COIL:

A choke coil, commonly known as choke, is an inductor used in electronic circuits.

Definition:

“A coil which consists of thick copper wire wound closely in a large number of turns over soft Iron laminated core is called a choke.”

Choke coil (or ballast) is a device that has high inductance and negligible resistance.

Explanation:

A choke coil, or simply a choke, is a type of inductor with a small resistance and a large inductance.

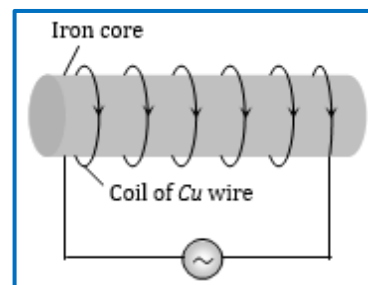
A choke coil reduces the current in AC circuit without consuming any power.

When resistance is used, the current reduces but energy losses occur due to heating. So, a choke coil is preferred.

- It is used to reduce the current in AC circuits without much loss of energy.
- It is also used in electrical circuits to block or "choke" high-frequency alternating current (AC) while allowing low-frequency AC to pass through.

In choke coil the inductance "L" of the coil should be large whereas resistance R is very small. Therefore, it consumes extremely small power.

The choke coil can be used only in ac circuits, not in dc circuits, because for dc ($\omega=0$) the inductive reactance " $X_L = \omega L$ " of the coil is zero, only the resistance of the coil remains effective which too is almost zero.

**Applications:**

Choke coils find applications in various electronic devices, including:

- 1) Power supplies for audio amplifiers
- (2) Radio frequency (RF) circuits
- (3) Other electronic systems where filtering and noise suppression are essential.

WORKED EXAMPLE:

20.4: Suppose we have an inductor in the form of a coil with an inductance of 0.02Henries (H) and connected in series with an AC signal having frequency of 50 Hz. Calculate the inductive reactance of coil.

RLC SERIES CIRCUITS:

An RLC circuit is an electrical circuit consisting of a resistor (R), inductor (L), and capacitor (C), connected in series or parallel.

RLC circuits are fundamental in electronics for filtering signals, tuning circuits, and resonant applications due to their ability to manipulate frequencies and impedances.

1. AC THROUGH RC SERIES CIRCUITS:**Definition:**

"The circuit in which a resistor (R) and a capacitor (C) are connected in series with AC source is called RC series circuit."

Explanation:

Consider a circuit containing resistor and capacitor connected in series with an alternating voltage source as shown in figure.

Due to series circuits same current will flow through each circuit element.

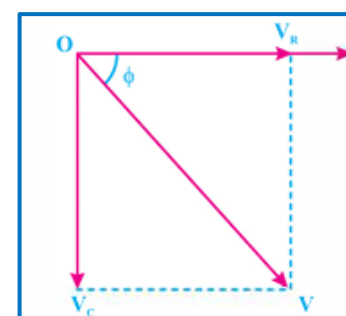
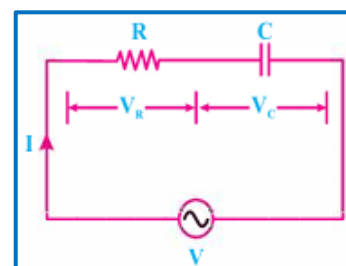
$$I = I_R = I_C$$

While voltage across resistor will be V_R and V_C across capacitor.

PHASOR DIAGRAM:

The phasor diagram is drawn taking current as reference direction as shown in figure.

In RC series circuits the voltage and current are in phase when considering resistor component of circuit.



And the voltage lags the current by $\pi/2$ radians in case of capacitor; it is drawn perpendicular to the current phasor in figure.

IMPEDANCE OF RC SERIES CIRCUIT:

From the phasor diagram, using Pythagoras theorem

$$V = \sqrt{V_R^2 + V_C^2} \dots\dots (1)$$

Substituting voltage across resistor, $V_R = I R$ and across capacitor, $V_C = I X_C$ in equation (1), we get

$$V = I \sqrt{R^2 + X_C^2}$$

Divided both sides by current I , we get

$$\frac{V}{I} = \sqrt{R^2 + X_C^2} \dots\dots (2)$$

Where $\frac{V}{I} = Z$ is the impedance of RC series circuit. i.e.

$$Z = \sqrt{R^2 + X_C^2}$$

Phase Angle:

Phase angle between current and voltage as shown in figure (Phasor diagram) is calculated by

$$\tan \phi = \frac{V_C}{V_R} = \frac{X_C}{R} \quad \text{Or} \quad \phi = \tan^{-1} \left(\frac{X_C}{R} \right)$$

2. AC THROUGH RL SERIES CIRCUITS:

Definition:

"The circuit in which a resistor (R) and an inductor (L) are connected in series with AC source is called RL series circuit."

Explanation:

Consider a circuit containing inductor and resistor connected in series with an alternating voltage as shown in figure.

Due to series circuits same current will flow through each circuit element.

$$I = I_R = I_L$$

The voltage across resistor and inductor is V_R and V_L respectively.

PHASOR DIAGRAM:

The phasor diagram is drawn taking current as reference direction as shown in figure.

When an inductor is connected with alternating voltage then due to generation of back EMF current lag the voltage by 90° , it is represented by line perpendicular to the reference phasor i.e., current I as shown in figure.

Since in resistor the current and voltage are in phase. In the phasor diagram it is V_R line.

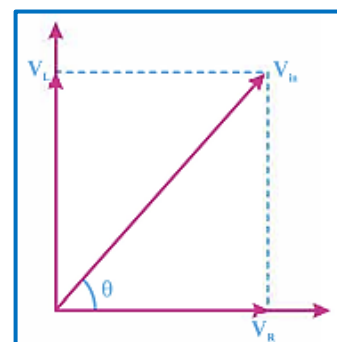
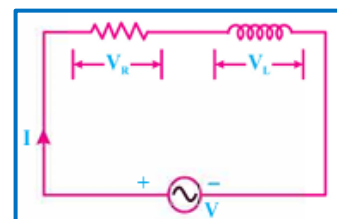
IMPEDANCE OF RC SERIES CIRCUIT:

To calculate the impedance of RL circuit, apply Pythagoras theorem on phasor diagram from figure.

From the phasor diagram, using Pythagoras theorem

$$V = \sqrt{V_R^2 + V_L^2} \dots\dots (1)$$

$$V = I \sqrt{R^2 + X_L^2}$$



Divided both sides by current I , we get

$$\frac{V}{I} = \sqrt{R^2 + X_L^2} \dots\dots (2)$$

Where $\frac{V}{I} = Z$ is the impedance of RC series circuit. i.e.

$$Z = \sqrt{R^2 + X_L^2}$$

Phase Angle:

Phase angle between current and voltage as shown in figure (b) is calculated by

$$\tan\phi = \frac{X_L}{R} \quad \text{Or} \quad \phi = \tan^{-1}\left(\frac{X_L}{R}\right)$$

WORKED EXAMPLE:

20.5: In RC circuit a capacitor of $50\mu\text{F}$ and resistor of 100Ω are connected in series with alternating voltage source of 220V and frequency of 50Hz . Find the (a) current in the circuit (b) phase angle between voltage and current.

3. RLC SERIES AC CIRCUIT:

Definition:

An RLC series circuit consists of a resistor (R), inductor (L), and capacitor (C) connected in series to an AC voltage source as shown in figure.

Explanation:

Since all three circuit elements are connected in series so same current will flow through them.

$$I = I_R = I_L = I_C$$

Hence to analyze the different properties of circuit we draw a phasor diagram of RLC series circuit as shown in figure.

Since the capacitive and reactance were determined as

$$X_C = \frac{1}{2\pi fC} \quad \text{and} \quad X_L = 2\pi fL$$

For smaller frequency capacitive reactance is much greater than inductive reactance i.e., $X_C > X_L$ causing the circuit to exhibit behavior as an RC circuit.

For larger frequency inductive reactance dominates over capacitive reactance i.e. $X_L > X_C$ and circuit behaves like an RL circuit. Hence the circuit is said to be more inductive than capacitive.

IMPEDANCE TRIANGLE:

Definition:

"The combined effect of resistance and reactance in RLC circuit is known as impedance."

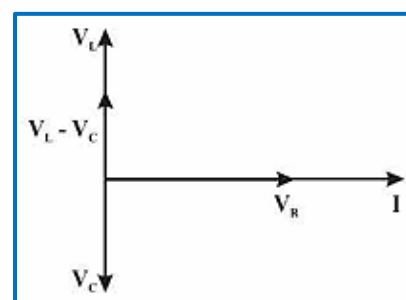
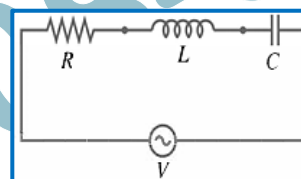
It is denoted by Z .

Explanation:

A circuit containing resistor, inductor and capacitor offer opposition to flow of current due to all these circuit elements known as impedance.

It is measured by the ratio of the rms value of the applied voltage to the rms value of resulting A.C.

$$Z = \frac{V_{rms}}{I_{rms}}$$



Mathematically:

The impedance triangle is shown in the figure. To draw an impedance triangle, represent the resistance (R) as the horizontal side of the triangle.

Represent the inductive reactance (X_L) or capacitive reactance (X_C) as the vertical side of the triangle.

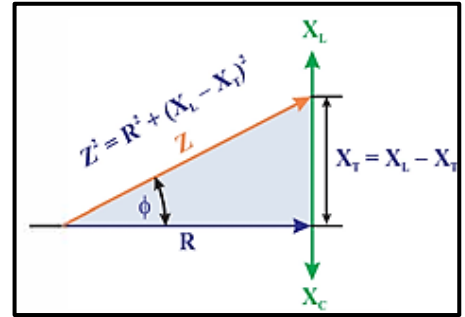
The direction (up or down) depends on whether it's inductive or capacitive reactance.

Now the hypotenuse of the triangle represents the impedance (Z).

From the right-angle triangle impedance can be calculated by using the following relation.

$$Z = \sqrt{R^2 + X_T^2} \dots\dots\dots (1)$$

Where Z is impedance, R is resistance in a circuit and X_T is the difference of capacitive and inductive reactance of circuit.

**POWER IN A.C CIRCUITS:**

The calculation of power dissipation in a resistor involves the formula $P = VI$. However, this equation is not applicable in circuits where capacitors and inductors are connected to an alternating voltage source.

The reason being that in a purely resistive circuit, the current and voltage are in phase. Whereas circuits containing capacitive or inductive elements exhibit a lead or lag of $\pi/2$ radians between the current and voltage, respectively, disrupting the applicability of this relationship.

Therefore, in such a case to determine power dissipation, project V onto the direction of reference current phasor as shown in figure.

In A.C. circuit the phase difference between applied voltage V and the current I is θ . The component of V along current I is $V \cos\theta$. Actually, it is this component of voltage vector which is in phase with current. So, the power dissipated in A.C circuit is $P = VI \cos\theta$

Where $\cos\theta$ represents power factor.

WORKED EXAMPLE:

20.6: A resistor (R) of 10 Ω and an inductor (L) of 0.05 H are connected in series. The AC voltage source has a peak voltage of 100 V and a frequency of 50 Hz. Find:

1. The impedance of the circuit.
2. The RMS current through the circuit.
3. The average power consumed by the circuit

SELF-ASSESSMENT QUESTIONS:

1. Compare and contrast the flow of AC through resistors versus capacitors in terms of phase shift.
2. Describe the conditions under which maximum power transfer occurs between a source and a load in an AC circuit.

RESONANT FREQUENCY:**Definition:**

For a certain frequency the capacitive and inductive reactance becomes equal, $X_C = X_L$. This frequency is called resonant frequency f_r and circuit is said to be in resonance state.

Calculation:

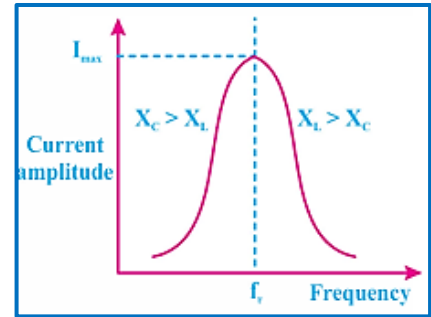
To calculate the resonant frequency, use the fact that

$$X_C = X_L$$

$$\frac{1}{2\pi fC} = 2\pi fL$$

Or
$$f_r = \frac{1}{2\pi\sqrt{LC}} \dots\dots (1)$$

From the figure, X_C and X_L are in opposite directions, therefore, at resonant frequency they cancel each effect in circuit. Now opposition to the current flow is solely offered by resistor, resulting in maximum current to flow through the circuit, moreover either side of resonant frequency the current in the circuit decreases.



USE OF RLC SERIES CIRCUIT:

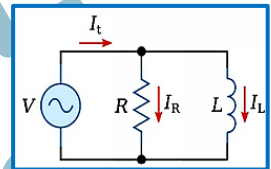
RLC series circuit is an important circuit it is used in; Electronic filters, Electromagnetic Interference (EMI) Suppression, Tuned Circuits, etc.

PARALLEL RLC CIRCUIT:

1. PARALLEL RL CIRCUIT:

Definition:

An RL parallel circuit is a setup where a resistor and an inductor are connected in parallel and share the same voltage source.



Explanation:

The RL parallel circuit is similar to the purely inductive parallel circuit. In parallel RL circuit, applied voltage “V” is same for both components and is therefore plotted in standard position in the phasor diagram.

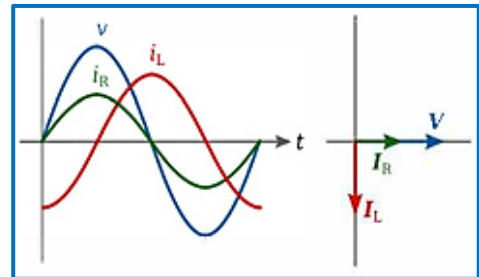
$$V = V_R = V_L$$

But the net current distributes in both components can be determined by the opposition (reactance) of the individual branches.

$$I = \sqrt{I_R^2 + I_L^2}$$

Wave and Phasor Diagram:

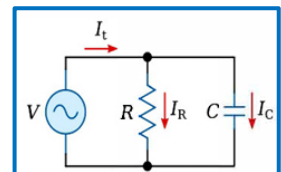
Figure shows a composite diagram of waveforms and phasors. Since the phasor diagram shows that the two branch currents are not in phase, it will be necessary to use phasor addition in order to determine the total current.



2. PARALLEL RC CIRCUIT:

Definition:

An RC parallel circuit (also known as an RC filter or RC network) is an electrical circuit consisting of a resistor R and a capacitor C connected in parallel, driven by a voltage source or current source.



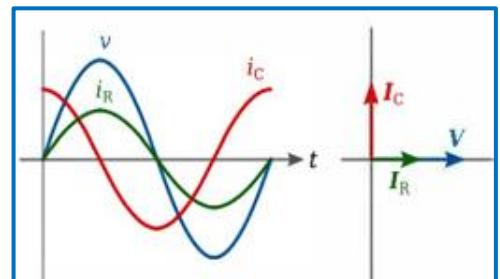
Explanation:

In a parallel RC circuit, the voltage is uniform across all components, while the total current is the sum of individual currents through the resistor and capacitor.

$$I = \sqrt{I_R^2 + I_C^2}$$

Wave and Phasor Diagram:

Figure shows a composite diagram of waveforms and phasors as per circuit conditions. The current phasors I_R and I_C are out of phase; therefore, phasor addition must be used to determine total current.



3. PARALLEL RLC CIRCUIT:

Definition:

A Parallel RLC AC Circuit is one where the resistor, inductor, and capacitor are connected in parallel to each other and the AC source.

Explanation:

In this configuration, the voltage across each component is the same, but the currents through them differ.

The parallel arrangement affects the overall impedance and current distribution in the circuit, making it distinct from its series counterpart.

In a parallel circuit, the voltage V (RMS) across each of the three elements remains the same. Hence, for convenience, the voltage may be taken as a reference phasor.

In the parallel RLC circuit, the supply voltage " V_s " is common to all three components, while the supply current " I_s " consists of three parts:

- 1) The current through the resistor (I_R)
- 2) The current through the inductor (I_L), and
- 3) The current through the capacitor (I_C).

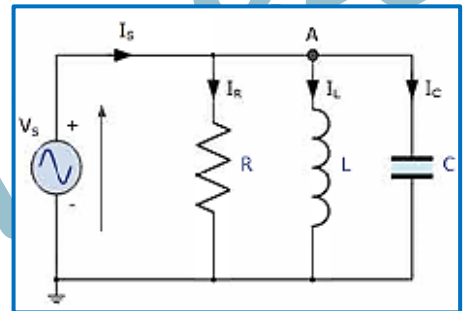
The current flowing through each branch, and therefore through each component, will differ from one another and from the supply current " I ".

The total current drawn from the supply is not simply the arithmetic sum of the three individual branch currents but their vector sum. i.e.

$$I = \sqrt{I_R^2 + (I_L - I_C)^2}$$

Similar to the series RLC circuit, we can solve this circuit using the phasor or vector method. However, in this case, the vector diagram will use the voltage as its reference, with the three current vectors plotted relative to this reference voltage.

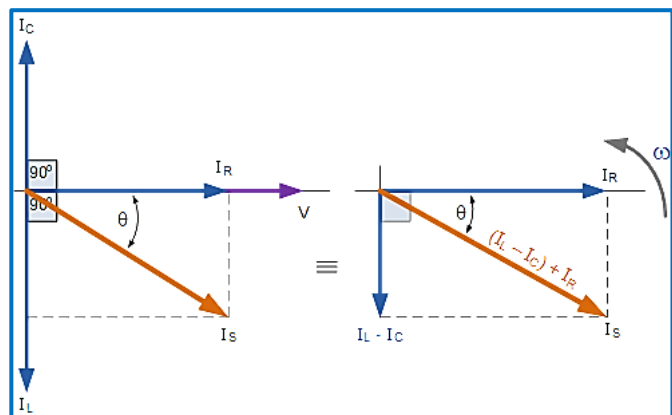
The phasor diagram for an AC RLC parallel circuit is created by combining the individual phasors for each component and adding the currents vectorially.



PHASOR DIAGRAM OF A PARALLEL RLC CIRCUIT:

From the phasor diagram of AC RLC parallel circuit, we observe that the current vectors form a right triangle, with the hypotenuse represented by " I_s ", the horizontal axis by I_R , and the vertical axis by $I_L - I_C$. This configuration form is known as a Current Triangle.

Consequently, we can apply Pythagoras's theorem to this current triangle to mathematically determine the individual magnitudes of the branch currents along the x-axis and y-axis. This will allow us to calculate the total supply current " I_s " of these components when the circuit's phase angle is given.



RESONANCE OF PARALLEL RLC AC CIRCUIT:

A parallel circuit containing a resistance, R , an inductance, L and a capacitance, C will produce a parallel resonance (also called anti-resonance) circuit when the resultant current through the parallel combination is in phase with the supply voltage.

At resonance there will be a large circulating current between the inductor and the capacitor due to the energy of the oscillations, then parallel circuits produce current resonance.

A parallel resonant circuit stores the circuit energy in the magnetic field of the inductor and the electric field of the capacitor. This energy is constantly being transferred back and forth between the inductor and the capacitor which results in zero current and energy being drawn from the supply.

In the solution of AC parallel resonance circuits, we know that the supply voltage is common for all branches, so this can be taken as our reference vector. Each parallel branch must be treated separately as with series circuits so that the total supply current taken by the parallel circuit is the vector addition of the individual branch currents.

$$I_R = \frac{V}{R}$$

$$I_L = \frac{V}{X_L} \quad I_L = \frac{V}{2\pi fL}$$

$$I_C = \frac{V}{X_C} \quad I_C = V \cdot 2\pi fC$$

Therefore,

$$I_T = \text{Vector sum of } (I_R + I_L + I_C)$$

$$I_T = \sqrt{I_R^2 + (I_L - I_C)^2}$$

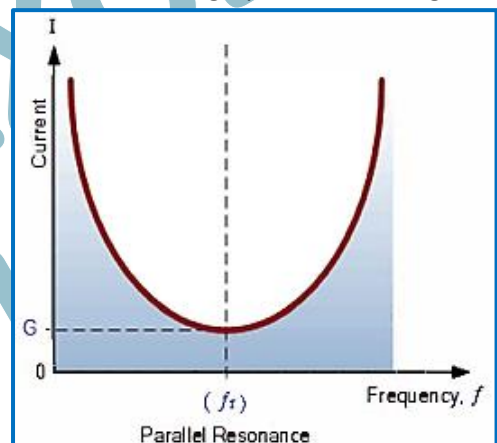
At resonance, currents I_L and I_C are equal and cancelling giving a net reactive current equal to zero. Then at resonance the above equation becomes.

$$I_T = \sqrt{I_R^2 + (0)^2} = I_R$$

Therefore, the circuit current at this frequency will be at its minimum value and the graph of current against frequency for a parallel resonance circuit is shown in figure.

The frequency response curve of a parallel resonance circuit shows that the magnitude of the current is a function of frequency and plotting this onto a graph shows us that the response starts at its maximum value, reaches its minimum value at the resonance frequency when $I_{\min} = IR$ and then increases again to maximum as “f” becomes infinite.

The result of this is that the magnitude of the current flowing through the inductor, “L” and the capacitor, “C” tank circuit can become many times larger than the supply current, even at resonance but as they are equal and at opposition (180° out-of-phase) they effectively cancel each other out.



As a parallel resonance circuit only functions on resonant frequency, this type of circuit is also known as “a Rejecter Circuit” because at resonance, the impedance of the circuit is at its maximum thereby suppressing or rejecting the current whose frequency is equal to its resonant frequency.

The effect of resonance in a parallel circuit is also called "current resonance".

SELF-ASSESSMENT QUESTIONS:

1. What factors determine the resonant frequency of an LC circuit?
2. How does the quality factor (Q-factor) relate to the sharpness of resonance in a circuit?

WORKED EXAMPLE:

20.7: An inductor of solenoid has a reactance of 500Ω is connected with an alternating voltage 220V and frequency of 50 Hz. Calculate the inductive reactance of solenoid.

MAXIMUM POWER TRANSFER THEOREM:

According to this theorem:

Definition:

“Maximum power is transferred from a source to a load when the impedance of the source matches with the impedance of the load”.

Explanation:

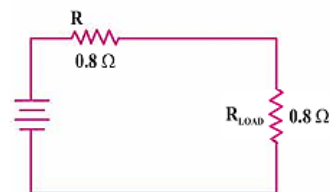
Since impedance is a measure of opposition a circuit presents to the flow of alternating current (AC).

Therefore, impedance includes both resistance and reactance, where reactance is the opposition due to capacitance or inductance.

When the impedance of the source matches the impedance of the load, the conditions for maximum power transfer are met. Under these circumstances, the electrical energy is efficiently transferred from the source to the load, resulting in maximum power transfer.

Mathematically, suppose R_L denotes the resistance of load and R_s is source resistance.

Since $R_L = R_s$ as shown in figure, maximum power transfer will occur. In practical terms, achieving maximum power transfer is often important in designing electronic circuits to optimize efficiency and minimize power loss.



WORKED EXAMPLE:

20.8: For an RLC circuit in parallel with a 5 Ω resistor, a 0.02 H inductor, and a 0.005 F capacitor, calculate the resonant frequency f_r ?

ELECTROMAGNETIC INDUCTION DEVICES:

1. METAL DETECTORS:

The oscillator circuit is used in metal detectors.

Metal detectors used for security checks operate on the principle of electromagnetic induction.

CONSTRUCTION & WORKING:

A simplified explanation of working of metal detector is described below:

1. Generating an Electromagnetic Field:

The metal detector contains a coil of wire through which an electric current flows, creating a magnetic field around the coil. This coil is often housed in a special arrangement, such as a loop or wand.

2. Interaction with Metals:

When a conductive metal object is brought into the vicinity of the electromagnetic field, it disturbs the field. This disturbance induces a secondary magnetic field in the metal object.

3. Eddy Currents:

The changing magnetic field induces circulating electric currents within the metal object, known as eddy currents. These eddy currents, in turn, generate their own magnetic fields.

4. Detection of Changes:

The metal detector has a receiver coil or coils that are in close proximity to the transmitter coil. The receiver coil(s) detects changes in the magnetic field caused by the presence of the metal.

5. Alert Mechanism:

When the metal detector senses a significant change in the magnetic field, indicating the presence of a metal object, it triggers an alert. This alert can be in the form of an audible sound, a visual signal, or both, depending on the design of the metal detector.

2. THE ELECTROCARDIOGRAPH:

The electrocardiograph (ECG) is a medical diagnostic tool used to record the electrical activity of the heart over a period of time.

PRINCIPLE:

The principle behind the electrocardiograph is based on the electrical signals generated by the heart during each heartbeat.

WORKING:

- The heart is a muscular organ that contracts rhythmically to pump blood throughout the body. This contraction is initiated and coordinated by electrical signals generated within the heart.
- Electrodes are attached to specific points on the skin of the patient. The standard placement involves attaching electrodes to the limbs and chest. These electrodes are conductive and are used to detect the electrical signals produced by the heart.
- The electrical signals produced by the heart are picked up by the electrodes. The detected electrical signals are amplified to make them more measurable and are then recorded on a graph or displayed on a monitor. The resulting graph is called an electrocardiogram.
- The ECG graph represents the electrical activity of the heart over time. It consists of waves and intervals, each of which corresponds to a specific phase of the cardiac cycle.
- Physicians analyze ECG to gather information about the heart's rhythm, rate, and various other aspects of its electrical activity.
- Deviations from the normal ECG pattern can indicate cardiac abnormalities, such as arrhythmias, ischemia, or other heart conditions.

3. OSCILLATOR CIRCUIT:**Definition:**

An oscillator circuit is an electronic circuit that generates a continuous periodic signal at a specific frequency.

Explanation:

Oscillator circuits produce carrier waves, which serve as the central frequency around which the information-carrying signal is modulated.

Oscillator circuits are designed to be tunable, allowing broadcasters to set the carrier frequency to a specific value. This tunability is essential for assigning unique frequencies to different radio stations, preventing interference between them. Oscillators are designed to be modulated by an information-carrying signal.

In amplitude modulation, the amplitude of the carrier wave varies according to the audio signal, while in frequency modulation; the frequency of the carrier wave is modulated. This modulation process allows the transmission of audio information.

The carrier wave itself does not convey information; it serves as a medium to carry the modulated information signal. The oscillator generates a stable carrier wave, ensuring that the modulated signal can be reliably transmitted over long distances. The carrier wave, when modulated with the audio signal, forms the composite radio wave that propagates through space. This transmitted signal can be received by radio receivers tuned to the carrier frequency.

In summary, the oscillator circuit is the heartbeat of a radio transmitter, generating the carrier wave that facilitates the transmission of modulated information signals.

Its stability, tunability, and modulation capabilities are pivotal for effective and interference-free radio broadcasting.

4. RESONANCE IN TUNING CIRCUIT OF RADIO:

In radio tuning circuits resonance is an important phenomenon that enables the selective reception of a desired radio frequency while rejecting others.

PRINCIPLE:

The principle of resonance is crucial in the tuning circuits of radio to enhance the reception of specific radio frequencies.

WORKING:

Resonance occurs when the inductive and capacitive reactance in a circuit cancel each other out, resulting in a condition where the circuit efficiently absorbs energy at a particular frequency.

Resonance is integral to the functioning of radio tuning circuits. It ensures that radios can selectively receive signals from desired stations with clarity and efficiency, making it possible to enjoy various broadcasted content without interference from other frequencies.

IMPORTANCE OF BROADCASTING:

- Broadcasting remains a powerful medium with far-reaching impacts on society.
- It informs, educates, entertains, and connects people, playing a crucial role in cultural preservation, economic development, and social cohesion.
- Its ability to reach a wide audience makes it an indispensable tool for communication in the modern world.

SUMMARY

- AC or alternating current refers to the electric charge flow that periodically reverses direction, typically sinusoidal.
- The frequency of AC current determines how many times per second the AC current reverses direction.
- Root Mean Square (RMS) equivalent steady DC voltage producing the same power in a resistive load as the AC voltage.
- Current and Voltage are in phase in a resistive load known as Ohm's Law for AC current as $I = \frac{V_{RMS}}{R}$.
- In pure capacitive circuit the current leads the voltage by 90 degrees or $\frac{\pi}{2}$ radians.
- In inductive circuit the current lags the voltage by 90 degrees or $\frac{\pi}{2}$ radians.
- The combined effect of resistance and reactance in RC or RLC circuits is known as impedance.
- Resonant frequency is the frequency at which an oscillating system naturally vibrates with the greatest amplitude. At resonant frequency the impedance of the system is at its minimum.
- The RLC series circuit configuration results in high impedance at resonance and a lower overall current compared to the individual component currents.
- The RLC parallel circuit configuration the components are connected in parallel branches, providing multiple current paths. At resonance, the impedance is at its minimum, allowing a higher overall current flow compared to the series circuit.
- Maximum power is transferred from a source to a load when the load impedance is equal to the source impedance.
- ECG is a medical device that records the electrical activity of the heart over time.
- Oscillator Circuit is an electronic circuit that produces a periodic, oscillating signal, often a sine wave or square wave.
- Choke Coil is an inductor used to block higher-frequency AC signals while allowing lower-frequency or DC signals to pass.
- Resonance is the condition in a circuit when the inductive reactance and capacitive reactance are equal in magnitude but opposite in phase, resulting in a maximum voltage or current at a particular frequency.

TEXTBOOK WORK

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. In alternating current, the flow of electric charge periodically reverses direction. The term for this specific phenomenon is:
(a) Alternation (b) Oscillation (c) Cycle (d) Resistance
2. The frequency of standard household alternating current in Pakistan is:
(a) 50 Hz (b) 60 Hz (c) 100 Hz (d) 120 Hz
3. The parameter for measuring the rate of change of alternating current or voltage with respect to time is:
(a) Amplitude (b) Frequency (c) Phase angle (d) Peak Voltage
4. In AC circuits, what is the phase difference between current and voltage in a purely resistive circuit?
(a) 0° (b) 45° (c) 90° (d) 180°
(2024) (MP:25) (2025)
5. A device used to measure the root mean square (RMS) value of an AC voltage or current is:
(a) Voltmeter (b) Oscilloscope (c) Ammeter (d) A VO meter
6. In an AC circuit. The term "reactance" represents the:
(a) Ohmic resistance only
(b) Resistance to DC currents only
(c) Capacitive and inductive opposition to current flow
(d) Total impedance, including resistance and phase angle
7. A type of circuit characterized by equal values of resistance and reactance is:
(a) Resistive (b) Inductive (c) Capacitive (d) Resonant
8. In an AC sine wave, the peak-to-peak voltage is:
(a) Double the peak voltage
(b) The difference between the peak voltage and zero
(c) The sum of the peak voltage and zero
(d) The difference between the positive and negative peak voltages
9. What is the power factor of a purely resistive circuit?
(a) 0 (b) 0.5 (c) 1 (d) -1
10. The purpose of a transformer in an AC power distribution system.
(a) To convert AC to DC (b) To step up or step-down voltage
(c) To store electrical energy (d) To regulate current flow

SECTION- B: SHORT ANSWERED QUESTIONS

1. Define RMS voltage and explain its significance in AC circuits.
2. Explain the difference between peak voltage, RMS voltage, and average voltage in AC circuits.
3. Define alternating current (AC) and explain how it differs from direct current (DC).
4. Define reactance in AC circuits and differentiate between capacitive and inductive reactance.
5. Describe the behavior of capacitors and inductors in AC circuits.
6. Explain the phase relationship between voltage and current in capacitive and inductive AC loads.
7. What is resonance in AC circuits? Discuss its conditions and applications.
8. Discuss the role of transformers in AC circuits and explain how they work.
9. Explain the behavior of an RLC series circuit in terms of impedance, resonance, and phase angle.

10. Describe the operation and applications of a transformer in AC circuits.

SECTION- C: LONG ANSWERED QUESTIONS

1. Explain the concept of phasor in the context of AC voltage. How are phasors used to represent sinusoidal voltages? (2024)
2. Describe the concept of impedance in AC circuits. How does impedance differ from resistance, and what are the units of impedance?
3. How does the reactance of an inductor and a capacitor change with frequency in an AC circuit? Provide an explanation based on the fundamental formulas?
4. Discuss the concept of resonance in RLC circuits. What conditions lead to resonance, and how does it affect the behavior of the circuit?
5. Compare and contrast series and parallel resonance in RLC circuits. What are the key differences between the two resonance configurations?
6. Explain the transient response of an RLC circuit when initially connected to an AC source. What happens to the currents and voltages in the circuit?
7. Why alternating current (AC) is commonly used for long-distance power transmission?
8. Describe the purpose and function of a choke coil in AC circuits.

SECTION- D: NUMERICAL

1. A resistor (R) of 20 ohms is connected in series with a capacitor (C) of $10\text{ }\mu\text{F}$ in an AC circuit with a frequency of 50 Hz. Calculate the total impedance? (3182 ohms) (2024)
2. For an inductor with an inductance (L) of 0.5 H and a frequency of 100 Hz, calculate the inductive reactance? (314.16 ohms)
3. In an RL circuit, the resistance (R) is 30 ohms, and the inductance (L) is 0.2 H. Calculate the total impedance at a frequency of 60 Hz? (81.1 ohms) (MP:25)
4. In an RC circuit, the resistance (R) is 50 ohms, and the capacitance (C) is $20\text{ }\mu\text{F}$. Calculate the capacitive reactance? (31.83 ohms)
5. An AC circuit has a resistance of 40 ohms, an inductive reactance of 30 ohms, and a capacitive reactance of 20 ohms. Draw the impedance triangle and calculate the total impedance. (41.2 ohms)
6. In a series RL circuit, the resistance (R) is 25 ohms, and the inductance (L) is 0.1 H. Calculate the phase angle and impedance at a frequency of 80 Hz? (63.8° , 502 ohms)
7. In a parallel RC circuit, the resistance (R) is 60 ohms, and the capacitance (C) is $30\text{ }\mu\text{F}$. Calculate the total current flowing through the circuit at a frequency of 120 Hz? (60 V)
8. In an RLC circuit, the resistance (R) is 50 ohms, the inductance (L) is 0.1 H, and the capacitance (C) is $50\text{ }\mu\text{F}$. Calculate the resonance frequency? (70.7Hz) (2025)

ASSIGNMENT: 20

SECTION- A: MULTIPLE CHOICE QUESTIONS

1. In purely resistive AC circuit, the phase difference between current and voltage is: (MP:25) (2025)
(a) 0° (b) 45° (c) 90° (d) 180°
2. The term 'reactance' represents the: (MP:25)
(a) Ohmic resistance only (b) resistance to direct currents only
(c) Total impedance (d) capacitive and inductive opposition to current flow
3. In AC circuits, what is the phase difference between current and voltage in a purely resistive circuit?
(a) 0° (b) 45° (c) 60° (d) 180° (2024)

SECTION- B: SHORT QUESTIONS

4. Explain the behavior of RLC series circuit in terms of impedance.
5. Explain the difference between peak voltage, RMS voltage and average voltage in AC circuits.

NUMERICAL PROBLEMS:

6. In an RLC circuit, the resistance (R) is 50 ohms, the inductance (L) is 0.1 H and the capacitance (C) is $50\mu\text{F}$. Calculate its resonance frequency. (2025)
7. In an RL circuit, the resistance (R) is 30Ω and the inductance (L) is 0.2H. Calculate the total impedance at a frequency of 60Hz. (MP:25)
8. A resistor (R) of 20 ohms is connected in series with a capacitor (C) of $10\mu\text{F}$ in an AC circuit with a frequency of 50 Hz. Calculate the total impedance? (2024)

SECTION- C: DETAILED QUESTIONS:

9. Explain the concept of phasor in the context of AC voltage. How are phasors used to represent sinusoidal voltages? (2024)
10. Discuss the relationship between voltage and current when AC passes through Capacitor. (With all relevant diagrams and equations)